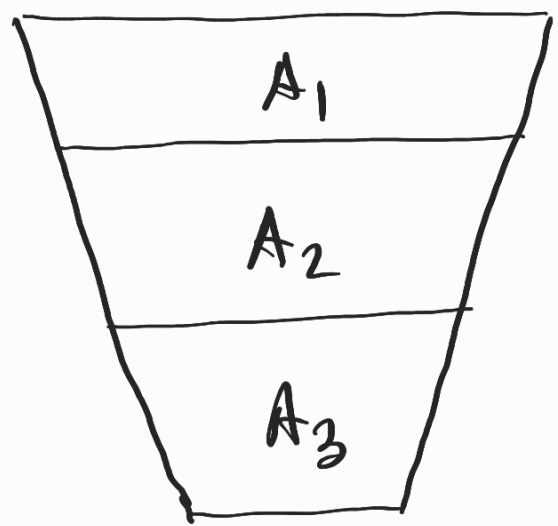
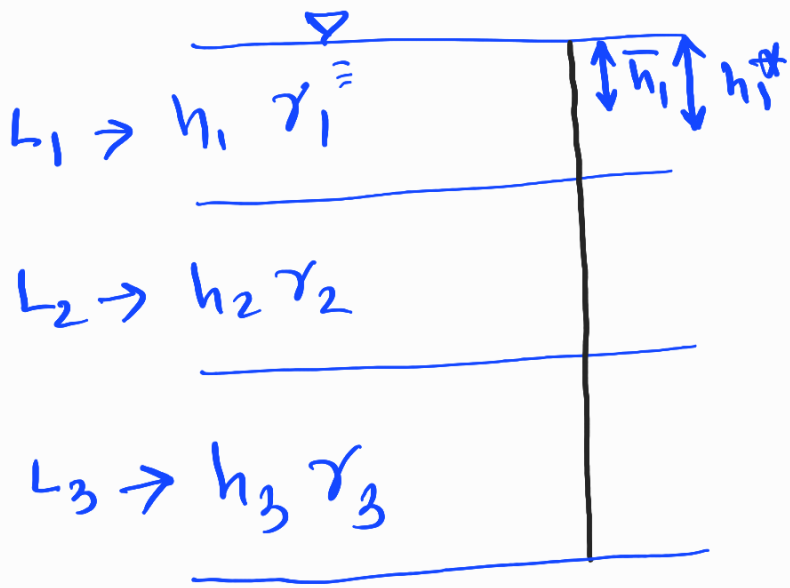


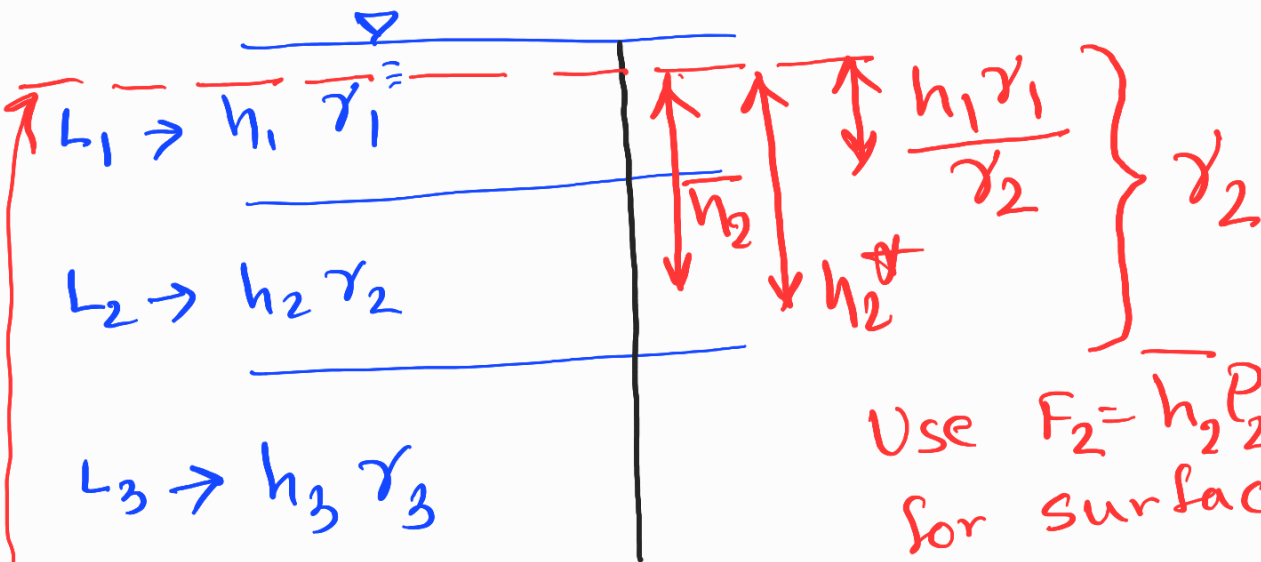


$L_1 \rightarrow$  directly use

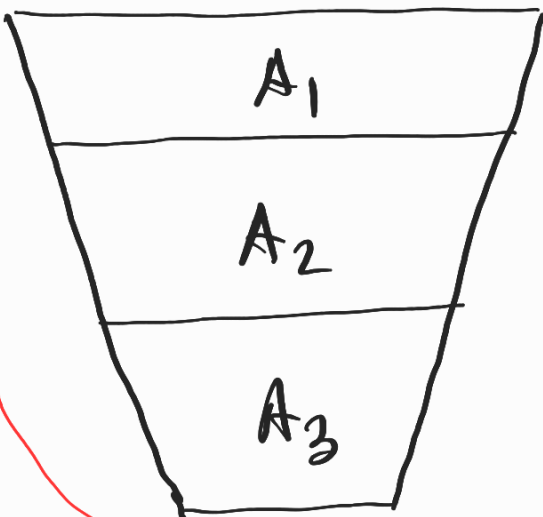
$$F_1 = \bar{h}_1 \rho_1 g A_1$$

and  $h_1^* = \bar{h}_1 + \frac{I_{c1} \sin^2 \theta}{A_1 \bar{h}_1}$

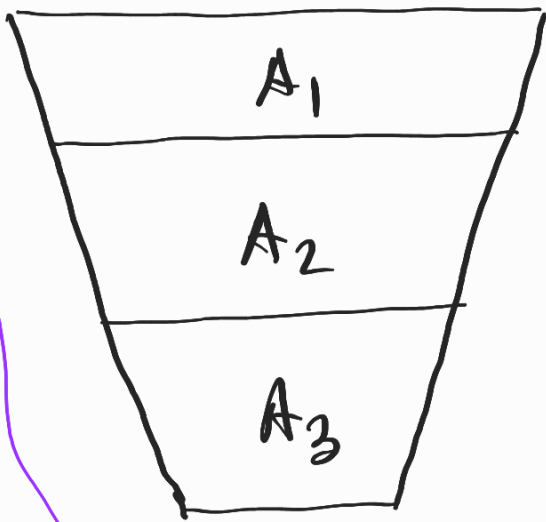
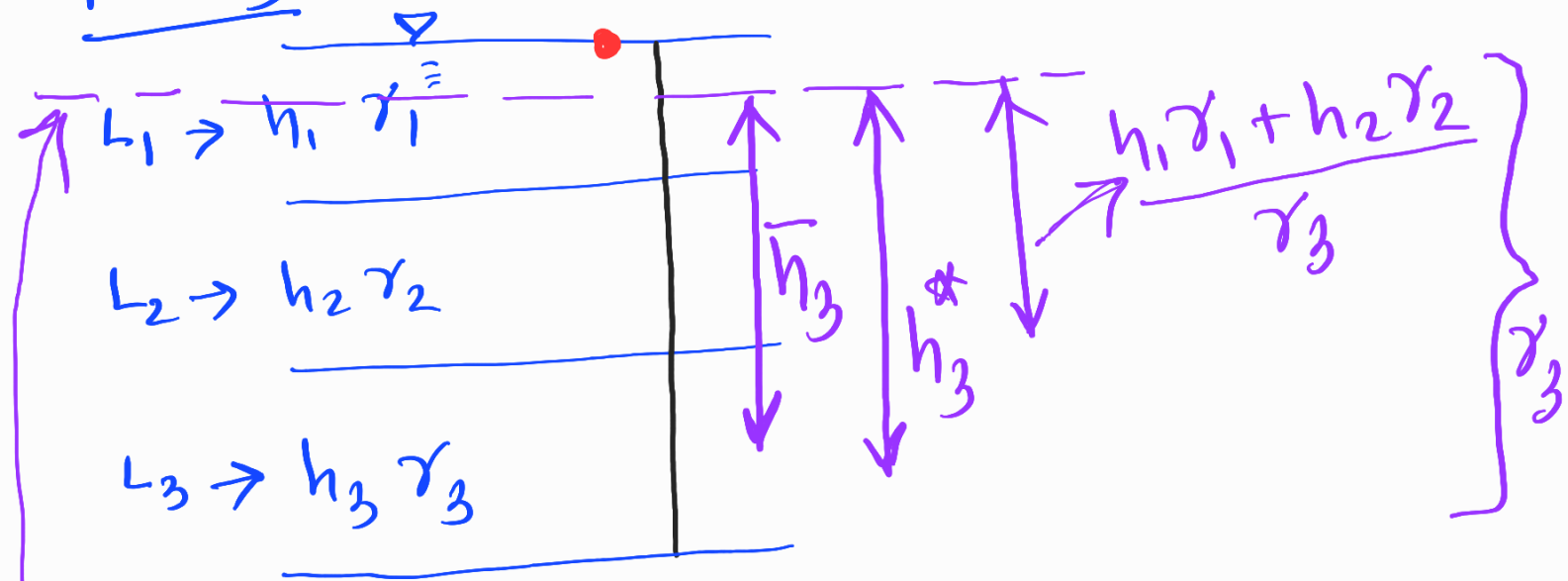
For  $L_2$



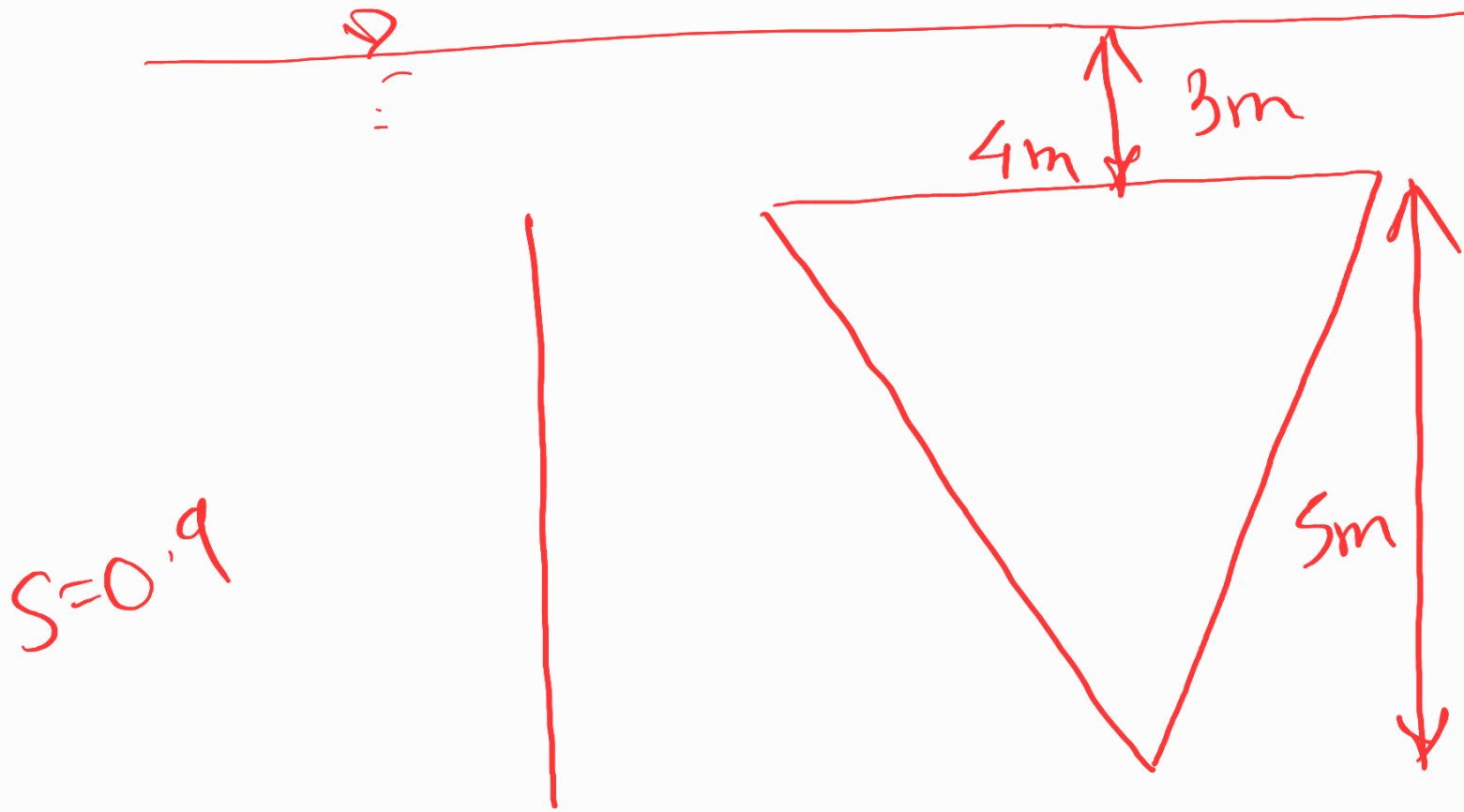
Use  $F_2 = \bar{h}_2 \rho_2 g A_2$   
 for surface in  
 contact with  
 second layer  
 with  $\bar{h}_2$  and  $h_2^*$   
 measured below  
 imaginary free  
 surface)



For  $L_3$



Use  $F_3 = \bar{h}_3 \rho_3 g A_3$   
 For the surface  
 in contact with  
 third layer  
 with  $h_3$  and  
 $h_3^*$  measured  
 below imaginary  
 free surface



$$F = \bar{h} \rho g A$$

$$= (3 + 5/3) \times 0.9 \times 9.81$$

$$\times \frac{1}{2} \times 4 \times 5$$

$$= 412 \text{ kN}$$

$$h^* = \overline{h} + \frac{I_{ca} \sin^2 \theta}{\overline{A h}}$$

$$= 4.67 + \frac{\frac{1}{36} \times 4 \times 5^3 \times \sin^2 90}{\frac{1}{2} \times 4 \times 5 \times 4.67}$$

$$= 4.97 \text{ m}$$